Your grade: 89.16%

Your latest: **89.16%** • Your highest: **89.16%** • To pass you need at least 80%. We keep your highest score.

coach

Almost there! Let's review your answers and then try again.

Your work showed skill in understanding autoencoders, GANs, GCN models, transformers, and several other advanced PyTorch techniques. However, there are areas where you can improve, particularly in preprocessing for recommendation systems, vision transformers, early stopping, PyTorch Lightning, SesameNet training, dimensionality reduction for word embeddings, and REST API setup. Before retrying this assignment, we recommend focusing on these key areas:

- **RecSys (Coding 2/4) - Model Training and Evaluation**: In Question 2, your understanding of the preprocessing steps for building a movie recommendation system was incomplete. Try reviewing this item.
- **Vision Transformers (VIT)**: In Question 5, the steps involved in developing and adapting a vision transformer for a specific dataset were not fully captured. Try reviewing this item.
- **Early Stopping (101)**: In Question 10, the importance of early stopping in training machine learning models was not fully understood. Try reviewing this item.
- **PyTorch Lightning (Coding)**: In Question 11, the functions that must be defined in a PyTorch Lightning module were not correctly identified. Try reviewing this item.
- **Semi-Supervised Learning (12) Modeling)**: In Question 13, the common issues encountered when running the training loop for SesameNet and their troubleshooting were not accurately addressed. Try reviewing this item.
- **Word Embeddings Intro (101)**: In Question 14, the techniques used to reduce the dimensionality of word embeddings for visualization were not correctly identified. Try reviewing this item.
- **API On-Premise with Deep Learning Model (Coding)**: In Question 20, the steps involved in setting up a model instance and creating a REST endpoint for predicting iris flowers were not fully captured. Try reviewing this item.

Good job on the rest of the assignment!

Retry assignment

1. In the context of training autoencoders, which of the following is a primary use case? — 1 / 1 point

- ○ Generating random noise
- ○ Calculating statistical summaries
- ◉ Data compression
- ○ Sorting data alphabetically

Correct
Correct! Autoencoders are primarily used for data compression by reducing the dimensionality of the data.

2. What are some important preprocessing steps for building a movie recommendation system? — 0.6 / 1 point

- ☑ Encoding categorical features

Correct
Correct! Encoding categorical features is an essential preprocessing step.

- ☑ Removing outliers from the dataset

This should not be selected
Incorrect. While removing outliers can be important, it is not always a necessary preprocessing step for recommendation systems.

- ☐ Adding Gaussian noise to the data
- ☑ Splitting the dataset into training and test sets

Correct
Correct! Splitting the dataset into training and test sets is crucial for evaluating the model's performance.

- ☐ Normalizing numerical features

3. What is the main purpose of the encoder component in an autoencoder? — 1 / 1 point

- ○ To reconstruct the input data from the latent space
- ○ To randomly shuffle the input data
- ◉ To compress the input data into a latent space representation
- ○ To classify the input data into predefined categories

Correct
Correct! The encoder compresses the input data into a latent space representation.

4. Which of the following are applications of Generative Adversarial Networks (GANs)? — 1 / 1 point

- ☑ Generating deep fake videos

Correct
Correct! GANs are often used to create deep fake videos, which can be both innovative and controversial.

- ☐ Reinforcing learning in autonomous agents
- ☑ Creating realistic images from textual descriptions

Correct
Correct! GANs can be used to generate realistic images from textual descriptions.

- ☐ Improving the speed of traditional convolutional neural networks

5. Which of the following steps is essential when setting up the training loop for a GAN? — 1 / 1 point

- ◉ Simultaneously training the generator and the discriminator
- ○ Ignoring the loss function of the generator
- ○ Using ReLU activation functions for both generator and discriminator
- ○ Only training the discriminator until it achieves 100% accuracy

Correct
Correct! Both the generator and the discriminator must be trained alternately to improve the GAN's performance.

6. When visualizing a graph using dimensionality reduction techniques, which of the following methods can be used to reduce the dimensionality of graph node features? — 1 / 1 point

- ○ K-Means Clustering
- ○ Random Forest
- ○ Linear Regression
- ◉ t-SNE (t-distributed Stochastic Neighbor Embedding)

Correct
Correct! t-SNE is a popular dimensionality reduction technique used for visualizing high-dimensional data.

7. Which of the following are important considerations when training and evaluating a GCN model? — 1 / 1 point

- ☑ Ensuring that the input features are standardized or normalized.

Correct
Correct! Standardizing or normalizing input features can improve the training process and model performance.

- ☑ Choosing an appropriate activation function.

Correct
Correct! The activation function can significantly impact the performance of the GCN model.

- ☐ Ignoring the impact of hyperparameter tuning.
- ☐ Using a fixed learning rate throughout training.
- ☑ Monitoring the training loss and validation loss.

Correct
Correct! Monitoring these losses helps to understand the model's performance and detect overfitting.

8. Which of the following are benefits of using transformers in various fields? — 1 / 1 point

- ☑ Effective parallelization during training

Correct
Correct! One of the significant benefits of transformers is their ability to be parallelized during training, leading to faster training times.

- ☑ Improved ability to capture long-range dependencies

Correct
Correct! Transformers are known for their ability to capture dependencies over long sequences, which is beneficial in many applications.

- ☑ Enhanced performance in tasks involving sequential data

Correct
Correct! Transformers have shown improved performance in tasks that involve sequential data, such as language translation and time series analysis.

- ☐ Reduced need for large datasets
- ☐ Simplified model architecture compared to traditional neural networks

9. What steps are involved in developing and adapting a vision transformer for a specific dataset? — 0.8 / 1 point

- ☐ Enhancing patches using GANs before feeding them into the transformer
- ☑ Partitioning images into fixed-size patches

Correct
Correct! Partitioning images into fixed-size patches is a crucial initial step in developing Vision Transformers.

- ☐ Training the model on the dataset with a suitable loss function
- ☐ Applying convolutional layers to extract features from patches
- ☑ Using positional encodings to represent the position of each patch

Correct
Correct! Positional encodings are used to retain the spatial information of image patches.

You didn't select all the correct answers

10. Why is early stopping important in training machine learning models? — 0.6 / 1 point

- ☑ It saves computational resources by halting training once further improvements are minimal.

Correct
Correct! Early stopping saves computational resources by stopping the training process when improvements become minimal.

- ☑ It simplifies the training loop by reducing the number of required epochs.

This should not be selected
Incorrect. While it may reduce the number of epochs, the primary goal is not to simplify the training loop.

- ☑ It prevents overfitting by stopping the training when the performance on a validation set starts to degrade.

Correct
Correct! Early stopping helps to prevent overfitting by monitoring performance on a validation set.

- ☐ It ensures the model reaches the absolute best performance.
- ☐ It helps in hyperparameter tuning by stopping trials that are not promising.

11. Which of the following functions must be defined in a PyTorch Lightning module? — 0.8333333333333334 / 1 point

- ☐ early_stop
- ☐ test_step
- ☑ validation_step

Correct
Correct! The validation_step function is used for defining the validation loop.

- ☑ training_step

Correct
Correct! The training_step function is essential for defining the training loop.

- ☑ configure_optimizers

Correct
Correct! The configure_optimizers function is crucial for defining the optimizer(s) to be used.

- ☐ model_step

You didn't select all the correct answers

12. In the context of self-supervised learning, what is the primary challenge when dealing with a dataset containing both labeled and unlabeled data? — 1 / 1 point

- ○ Removing noise from the labeled dataset.
- ○ Ensuring that the labeled data is completely accurate.
- ○ Generating high-quality labels for the labeled data.
- ◉ Balancing the contributions of labeled and unlabeled data during training.

Correct
Correct! One of the main challenges is to effectively balance the contributions from both labeled and unlabeled data.

13. What are the common issues you might encounter when running the training loop for SesameNet and how can they be troubleshooted? — 0.6 / 1 point

- ☐ Training accuracy can be ignored as long as validation accuracy is high.
- ☑ Loss function divergence can be fixed by increasing the learning rate.

This should not be selected
Incorrect. Increasing the learning rate can actually worsen the problem. Instead, try lowering the learning rate or checking for other issues.

- ☑ Data loader performance issues can be addressed by using parallel data loading.

Correct
Correct! Parallel data loading can enhance the performance of data loaders.

- ☐ Code bugs can be resolved by thorough debugging and testing.
- ☑ Model overfitting can be mitigated by using dropout.

Correct
Correct! Dropout is a common technique to prevent overfitting in neural networks.

14. Which of the following techniques are used to reduce the dimensionality of word embeddings for visualization? — 0.8 / 1 point

- ☐ Histogram Equalization
- ☑ TSNE

Correct
Correct! TSNE is commonly used for reducing the dimensionality of word embeddings for visualization purposes.

- ☐ Linear Discriminant Analysis (LDA)
- ☑ Principal Component Analysis (PCA)

Correct
Correct! PCA is another technique used to reduce dimensionality of word embeddings.

- ☐ Fourier Transform

You didn't select all the correct answers

15. What is a key benefit of using pretrained models from Hugging Face? — 1 / 1 point

- ○ They automatically correct grammatical errors in text
- ○ They are specifically designed for image processing
- ◉ They reduce the time and resources needed to train models from scratch
- ○ They provide a graphical user interface for model training

Correct
Correct! Pretrained models save significant time and resources compared to training models from scratch.

16. What is the main purpose of using Glovie word embeddings in Natural Language Processing? — 1 / 1 point

- ○ To perform spell check on a document
- ◉ To convert words into numerical representations based on their co-occurrence statistics
- ○ To sort words alphabetically in a corpus
- ○ To translate words from one language to another

Correct
Correct! Glovie embeddings convert words into numerical representations based on their co-occurrence statistics, which helps in capturing the meaning of words.

17. Which of the following are key aspects that drive the performance of Inception Networks? — 1 / 1 point

- ☐ Elimination of fully connected layers.
- ☑ Incorporation of 1x1 convolutions to reduce dimensionality.

Correct
Correct! 1x1 convolutions are used in Inception Networks to reduce dimensionality and computational cost.

- ☑ Use of multiple filter sizes in each module.

Correct
Correct! Inception Networks leverage multiple filter sizes within each module to capture various aspects of the data.

- ☑ Efficient grid size reduction.

Correct
Correct! Efficient grid size reduction helps in maintaining computational efficiency while processing data at multiple scales.

- ☐ Utilizing only one type of activation function.

18. In the context of Convolutional Neural Networks (CNNs), what is the primary purpose of implementing forward hooks? — 1 / 1 point

- ○ To perform data augmentation on the input images
- ○ To visualize the gradient flow during the backward pass
- ○ To modify the learning rate dynamically during training
- ◉ To capture and inspect intermediate activations during the forward pass

Correct
Correct! Forward hooks allow you to capture and inspect intermediate activations, which can provide valuable insights into the network's behavior.

19. Which tool can be used to test a REST API by sending various types of HTTP requests? — 1 / 1 point

- ○ Nginx
- ○ MySQL
- ○ Docker
- ◉ Postman

Correct
Correct! Postman is widely used for testing REST APIs by sending different types of HTTP requests.

20. Which of the following are steps involved in setting up a model instance and creating a REST endpoint for predicting iris flowers? — 0.6 / 1 point

- ☐ Define database schema
- ☑ Set up cloud storage

This should not be selected
Incorrect. While setting up cloud storage is vital, it's not a direct step in creating a model instance or REST endpoint for iris flower predictions.

- ☑ Load model parameters

Correct
Correct! Loading model parameters is a crucial step in setting up a model instance.

- ☐ Run the server
- ☑ Create a REST endpoint

Correct
Correct! Creating a REST endpoint is necessary for making the model accessible for predictions.